



Final Exam (60 Points) - Model (2)

Q1: [Multiple Choice] Identify the letter of the choice that best answers the question

[36 x 1.0 = 36 points]

1. RTN is a way to describe the microoperations.
☒ TRUE (B) FALSE A 9
2. Register that contains the opcode instruction is ____
(A) MBR (B) MAR (C) PC ☒ IR
3. What is the minimum negative integer in one byte? ____
☒ -128 (B) -127 (C) -255 (D) -256
4. Architecture indicates its performance while organization indicates its hardware.
(A) TRUE ☒ FALSE
5. Register that specifies the address in memory is ____
(A) MBR ☒ MAR (C) PC (D) IR
6. Intel and AMD does not use the same architecture.
(A) TRUE ☒ FALSE
7. In IEEE double-precision representations, the exponent part occupies ____ bits.
(A) 8 (B) 23 ☒ 11 (D) 52
8. In IEEE single-precision representations, the exponent has a bias of ____
(A) 128 ☒ 127 (C) 1023 (D) 1024
9. 2's complement representation has two representations of zero.
(A) TRUE ☒ FALSE
10. ANSI code uses ____ bits to represent each character.
(A) 7 ☒ 8 (C) 16 (D) 32
11. Processor that uses little-endian is optimized for division instruction.
☒ TRUE (B) FALSE
12. 1's complement representation of "10000001" refers to decimal value ____
(A) -1 (B) 65 ☒ -126 (D) -127
13. In IEEE 32-bit representations, the fractional part of the mantissa occupies ____ bits.
(A) 8 ☒ 23 (C) 11 (D) 52
14. Bus interface unit: executes instruction, while execute unit: fetches the instruction.
(A) TRUE ☒ FALSE
15. What is the register size in processor 8086 ____
(A) 8 ☒ 16 (C) 32 (D) 64
16. In BIU, while instruction is executed, other instructions are fetched in advance and stored in ____
(A) ALU ☒ Instruction queue (C) Flag (D) Instruction decoder
17. In 8086, each memory segment size is up to ____ bytes
☒ 64k (B) 32k (C) 64 (D) 32
18. As one goes down the memory hierarchy from Registers to Magnetic tap, the following conditions are occur except ____
☒ Increasing cost per bit (B) Increasing capacity (C) Increasing access time (D) Decreasing frequency of access of the memory by the processor
19. The L1 cache is slower than the L3 cache.
(A) TRUE ☒ FALSE
20. When the processor attempts to read a word of memory, a check is made to determine if the word is in the cache.
☒ TRUE ☒ FALSE
21. ____ refers to whether memory is internal or external to the computer.
☒ Location (B) Access (C) Hierarchy (D) Tag
22. For random-access memory, ____ is the time from the instant that an address is presented to the memory to the instant that data have been stored or made available for use.
(A) Memory cycle time (B) Direct access (C) Transfer rate ☒ Access time
23. No single technology is optimal in satisfying the memory requirements for a computer system.
☒ TRUE (B) FALSE

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24. A portion of main memory used as a buffer to hold data temporarily that is to be read out to disk is referred to as a —
 (A) Disk cache (B) Latency (C) Virtual address (D) Miss
25. Fully associative cache is expensive to implement.
 (A) TRUE (B) FALSE
26. The performance of cache memory is frequently measured in terms of a quantity called —
 (A) cache miss (B) cache hit (C) hit ratio (D) miss ratio
27. Replace that block in the set that has been in the cache longest with no reference to it. This refers to —
 (A) MFU (B) LRU (C) FIFO (D) LFU
28. Direct mapping; each block maps into any line of the cache.
 (A) TRUE (B) FALSE
29. k-way means Block can be found in k-line in a set.
 (A) TRUE (B) FALSE
30. Interrupt cycle is executed — next instruction fetch.
 (A) before (B) after (C) with
31. Direct addressing requires two memory accesses.
 (A) TRUE (B) FALSE
32. The instruction is decoded to determine what action is required, this refers to —
 (A) Fetch instruction (B) Interpret instruction (C) Fetch data (D) Process data
33. — is used for debugging.
 (A) Direction flag (B) Trap flag (C) Interrupt flag (D) Overflow flag
34. A data hazard called — occurs if the read takes place before the write operation is complete.
 (A) RAW (B) WAR (C) WAW
35. The integer part of the mantissa need to be stored.
 (A) TRUE (B) FALSE
36. In two-stage pipeline, while the second stage is executing the instruction, the first stage can fetch and buffer the next instruction. This is called —
 (A) prefetch (B) postfetch (C) preexecute (D) postexecute

Q2: [Multiple Choice] Identify the letter of the choice that best answers the question [12 x 2.0 = 24 points]

27. A logical cache stores data using —
 (A) Physical addresses (B) Virtual addresses
28. Convert decimal number 3.5 into IEEE-754 32-bit floating-point representation is —
 (A) 4060-0000H (B) C060-0000H (C) 4000-0000H (D) C000-0000H
29. In IEEE-754 32-bit floating-point, hexadecimal value "4000-0000" represent decimal value —
 (A) -1 (B) 1 (C) 1.5 (D) 2
30. Overflow can be detected by the — operation between carry-out of MSB and carry-in to MSB.
 (A) AND (B) XOR (C) OR (D) XNOR

If the RAM has a two byte data word and address bits are grouped for direct mapping with a cache memory as follows: Tag=14 bits, Line=9 bits, Word=5 bits.

31. How many blocks does the RAM contains?
 (A) 4M (B) 8M (C) 16M (D) 32M
32. What is the number of lines in the related cache?
 (A) 128 (B) 256 (C) 512 (D) 1K
33. What is the size of RAM in bytes?
 (A) 128MB (B) 256MB (C) 512MB (D) 1GB
34. How many blocks allocate to the same cache line?
 (A) 2k (B) 4k (C) 8k (D) 16k
35. What is the size of one block in bytes?
 (A) 64 (B) 128 (C) 256 (D) 512

For the hexadecimal main memory address 111111, show the following information, in hexadecimal format:

36. Using a associative cache format [Tag=22 bits, Word=2 bits], Tag/Word = —
 (A) 44444 / 1 (B) 44444 / 2 (C) 11111 / 4 (D) 41114 / 1
37. Using a two-way set-associative cache format [Tag=9 bits, Set=13 bits, Word=2 bits], Tag/Set/Word = —
 (A) 44 / 444 / 1 (B) 22 / 444 / 4 (C) 22 / 444 / 1 (D) 24 / 444 / 1
38. Using a direct-mapped cache format [Tag=8 bits, Line=14 bits, Word=2 bits], Tag/Line/Word = —
 (A) 22 / 444 / 2 (B) 11 / 444 / 1 (C) 11 / 555 / 1 (D) 33 / 444 / 3

Best Wishes
 Dr. Muhammad Haggag